

Car Setup Guide

Richard Burns Rally has one of the most detailed car-setup systems ever put in a rally simulation. Plenty of drivers are happy with the standard setup, but if you want to explore the finer points of rally driving, tuning is essential. This guide walks through the setup menu, which you can open before each event or stage.

The default setups were made by Richard Burns together with an engineer he worked closely with. If you decide to go down the rally-engineering path, expect to lose hours: changing one value affects many others, and together they change how the car behaves.

NOTE This is a plain-English edit of a community guide that had been machine-translated. Parameter names follow standard rally-setup terms and may differ slightly from the labels in the in-game menu.

Setup screen controls

On the setup screen the controls differ slightly from the standard ones.

1. Use the **Up** / **Down** arrows to highlight a value.
2. Press **Enter** to activate or deactivate user-defined values.
3. When the arrows appear, use **Left** / **Right** to edit the value.

IMPORTANT Press Enter before and after adjusting a value with the left and right cursors.

Differential

The differential settings change the minimum and maximum lock of each differential in the car.

What a differential does

A differential does three jobs:

1. It transmits the engine's power to the wheels.
2. It lets the wheels at each end of an axle turn at different speeds.
3. It acts as the final drive, reducing rotational speed one last time before power reaches the wheels.

Every differential in the car ranges from 0% lock (free rotation) to 100% lock (fully locked). "Lock" is the percentage to which a differential is locked under throttle, brake and gear. When a differential is locked, both wheels on that axle turn at exactly the same speed.

Differential maps

Differential maps let you adjust lock pressure with a gradual progression from 0 to 100% in relation to throttle, brake and overall speed. They control how the car's three differentials (front, centre and rear) behave.

NOTE The ECU (engine control unit) always takes the highest lock value from throttle, brake and gear, and uses that.

IMPORTANT Never set differential map values to zero. See the caveat at the end of this section.

Central differential

This map is used for conventional driving and driving in a straight line.

Change	Effect on balance	Other consequences
▼ Decrease (less lock)	More oversteer	Less stable under braking
▲ Increase (more lock)	More understeer and loss of stability	More stable under braking

Central map, left-foot brake

If you use the brake and throttle at the same time, the ECU switches off the central differential map and uses this one instead. This technique is known as left-foot braking.

Change	Effect on balance	Other consequences
▼ Decrease	More oversteer	Less stable under braking
▲ Increase	More understeer and loss of stability	More stable under braking

IMPORTANT For advanced drivers.

ADVICE When left-foot braking, keep these values lower than the central differential values. It helps rotate the car into a corner.

TIP Use a controller setup that lets you press the throttle and brake at the same time.

Front differential

Change	Positive	Negative
▼ Decrease	The car turns in more easily on corner entry	Less traction out of corners
▲ Increase	More traction on corner exit	Harder to turn the car in at corner entry (more locked), and more steering effort on exit

ADVICE Use a lower lock percentage on tarmac stages than on gravel.

Rear differential

Change	Positive	Negative
▼ Decrease	A very slight increase in high-speed stability	Loss of traction
▲ Increase	More traction	Understeer in tight turns and loss of stability

ADVICE As a general rule, keep these values fairly high. For example, if 50% throttle is mapped to 25% lock and the differential torque is 600 Nm, the actual torque through the differential in that state is 150 Nm.

Differential torque

Differential torque is the adjustment from the minimum to the maximum lock pressure of a differential, measured in newton-metres (Nm).

NOTE These values are tuned for each car and each stage. If you think they should be higher, you can adjust them in the differential maps.

Maximum central differential torque

Change	Initial effect	Other consequences
▼ Decrease	Stronger initial response, more sensitive steering	Increases the chance of a rollover
▲ Increase	More stable	Increases the chance of understeer

Maximum front differential torque

Change	Initial effect	Other consequences
▼ Decrease	Less traction on the front wheels	Easier handling in tight turns
▲ Increase	More traction on the front wheels	Harder to control the exact direction

Maximum rear differential torque

Change	Initial effect	Other consequences
▼ Decrease	Less traction	Helps prevent oversteer
▲ Increase	More traction out of corners through the rear wheels	Can cause some oversteer

Handbrake

As well as applying the rear brakes, the handbrake disconnects drive from the engine to the rear wheels by opening the central differential.

The percentage is how much control pressure is needed to open the central differential, which helps you pull off a clean handbrake turn.

Handbrake limit

Again a percentage of the control input needed, this time to reach the handbrake's differential map in place of the central differential map.

Caveat: don't zero out differential values

In RBR NGP physics, the differential maps (**FrontDiffThrottle** , **RearDiffThrottle**) and the speed scale (**CenterSpeedMapFactor**) control how the differentials lock under throttle, braking and speed. They only work if they are filled in.

If this is zeroed	What happens
Front and rear differential map values	The progression curves lock up or break. The differentials end up open or unpredictable as speed and throttle increase.
CenterSpeedMapFactor set to 0.000000	Speed-sensitive unlocking of the centre differential is switched off. It feels numb and bound up in high-speed sweeps, and unpredictable on power exit.

Without active differential curves, NGP falls back to flat or open behaviour. That kills power transfer out of slides and makes the handling snap.

WHY SETUPS END UP ZEROED

- **Incomplete manual edits.** Copying setup blocks, or converting legacy files, without filling in all 10 index parameters leaves placeholder zeros across the array.
- **Tool or plugin corruption.** Saving setups through third-party tools or mismatched NGP versions can fail to write the lookup curves, which clears the active differential data.
- **Failed “locked diff” hacks.** Setting `CenterSpeedMapFactor` to zero is an attempt to force a fixed centre differential. It actually removes the speed-sensitive unlocking, so the car feels bound up in high-speed sweeps and unpredictable on power exit.

ADVICE Stick to setups with fully populated differential maps.

Suspension

Before you start

This menu lets you change how the car rides. For both rough and smooth surfaces you can adjust the dampers and springs to get a good balance. On narrower, twistier stages you will also need to adjust the geometry and the anti-roll bars.

NOTE Any change to the suspension must be the same on both sides of the car. If you change the rebound on the front left, make the same change on the front right.

DAMPERS

Dampers absorb (dampen) the wheel's energy when the car goes over a bump. They also control how quickly the wheel returns to the road, which would otherwise depend only on how the springs extend. All figures are in kN/m/s.

Compression

The energy absorbed inside the damper when the car goes over a bump or the edge of the road. Increase the value and the car feels stiffer; decrease it and the car feels softer.

TIP Set compression too high and the car tends to bounce off the road. Set it too low and the damper will not absorb the energy when the car goes over a pothole.

Rebound

Controls how fast the damper extends after compression, returning the tyre to the road.

ADVICE The lower the value, the more traction the car has. Too low and the car tends to bounce.

Fast compression

Works like compression and cushions the pits, potholes and bumps you meet on a stage. It has nothing to do with car speed. It describes how fast the damper compresses after a significant impact.

Fast compression limit

The speed (in metres per second) at which your fast compression settings take over from your normal compression settings.

TIP Too high and you may never reach the limit. Too low and you will always be above it.

SPRINGS

The suspension has two springs. The main spring does most of the work under load. The auxiliary spring comes into play when the wheel is lightly loaded, to give extra traction, or when it is unloaded, to make sure the damper extends fully.

Spring length

Changes the height of the chassis above the ground and the length of the suspension travel.

TIP There are simpler ways to change ride height: use the ride height setting in the Geometry menu. For car setups, leaving the spring length at its default is recommended.

Spring stiffness

The force needed to compress the spring. The higher the number, the harder the spring; a lower number gives a softer spring. This affects how the car handles over potholes.

ADVICE Stiffer springs help on higher-speed stages. In mud or snow, softer springs (a smaller number) can give you an advantage.

NOTE Auxiliary spring length and stiffness are tuned for each car, so leave them at their defaults.

Geometry (front and rear)

About geometry

The geometry menus change the wheel alignment and the height of the chassis above the ground. They apply to the front and rear of the car.

Top mount position

The mounting at the top of the axle, where it attaches to the body. Move it forwards or backwards to change the pivot advance (caster).

ADVICE The more negative the pivot advance, the better the car turns into corners. It can also increase loss of direction over potholes.

Track rod length

Adjusting the track rod gives the car more or less toe-in.

Ride height

Adjusts the height of the chassis above the ground. Adjusting the front relative to the rear also changes the weight distribution and the tilt of the car. Raising the rear moves the centre of gravity forwards.

TIP Raising the chassis gives more suspension travel and ground clearance, so the car takes less damage. Lowering it keeps the centre of gravity low, which is more stable.

Camber

Adjusts the wheel's lean angle.

ADVICE Change the camber to suit the road surface and to make sure the tyre wears evenly across its whole tread.

Anti-roll bars and steering

About anti-roll bars and steering

The anti-roll bar controls how much the body rolls between the left and right wheels. Steering is the

relationship between your controller's steering input and the car's maximum lock.

Maximum steering lock

With the value at MAX and full steering input, the car's wheels physically turn more. With the value at minimum and full steering input, the wheels turn less.

Front anti-roll bar stiffness

Affects how quickly the car's weight moves across the front axle, between the two front wheels. The higher the number, the stiffer the bar and the slower the weight transfer.

TIP A lower value gives a quicker initial turn-in. A higher value gives the car more stability at high speed.

Rear anti-roll bar stiffness

Affects how quickly the car's weight moves across the rear axle, between the two rear wheels. The higher the number, the stiffer the bar and the slower the weight transfer.

ADVICE Lower (softer) values give more traction but can cost stability.

ADVICE When it is wet, it can help to set both front and rear bars softer.

Tyres

Pirelli tyres

As a tyre goes through braking, acceleration and cornering, its pressure rises as the air inside heats up and expands. The pressure in the menus is for a cold tyre.

TIP On long, hot stages you may benefit from a slightly lower tyre pressure.

Brakes

About brakes

This section lets you change the balance between the front and rear brakes and the total maximum braking pressure.

Maximum brake pressure

The maximum braking pressure at the wheels. Increase it too much and the wheels can lock and the car slides. On a loose, wet or icy surface you need less force to start a wheel lock.

TIP If you are happy with the car's balance, keep the front-to-rear brake distribution the same. Set the rear brake too low and the car can slide off the road when braking. Set it too high and the rear tends to lock up under braking.

Brake pressure distribution

The percentage of the total braking force applied to that pair of wheels. The total always equals 1 (100%).

Brake disc diameter

The size of the brake disc.

Gearbox

About the gearbox

This section lets you change the gearbox ratios for a higher top speed (with less acceleration) or a lower top speed (with more acceleration). Which you choose depends on whether the stage is fast or twisty with tight corners.

Gear protector

Keeps the gearbox safe by shifting down in situations where the engine could be damaged.

Gear ratios

The ratio set changes the car's overall gear ratios to suit different types of stage.

Setting	Top speed	Acceleration
Long	Maximum top speed	Low
Medium	Average	Average
Short	Lowest top speed	Maximum

TIP If you run at the rev limit for a long time, you may need longer ratios: choose Medium or Long. If you never reach maximum revs in top gear, try a shorter ratio.